Summary Tables of `MAXCUT` Data

1 Tables for depth $P = 1$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	-	-	-
TQA*	-	-	-	-
		SV		
Fourier	61 [10–20]	10 [12]	699 [10–20]	570 [10–20]
Fixed Angle	55 [10–20]	-	-	289 [10–20]
Fixed Angle*	55 [10–20]	-	-	273 [10–20]
Interp	83 [10–20]	10 [12]	805 [10–22]	382 [10–20]
Linear Ramp*	-	-	-	-
TQA	65 [10–20]	-	592 [10–17]	280 [10–20]
TQA*	63 [10–20]	-	584 [10–17]	60 [10–20]
Recursive TS	81 [10–20]	10 [12]	481 [10–22]	368 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	81.1 ± 1.2	75.7 ± 1.5	71.5 ± 2.1	76.3 ± 1.4
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Interp (Aer)	81.1 ± 1.2	75.7 ± 1.5	71.5 ± 2.1	76.3 ± 1.4
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6
Linear Ramp*	-	-	-	-
Recursive TS	-	74.4	-	75.2 ± 0.3
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1
Fixed Angle	78.4 ± 2.6	-	-	80.7 ± 3.1
Fixed Angle*	79.1 ± 2.9	-	-	81.7 ± 4.3
Interp	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2
Linear Ramp*	-	-	-	-
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9
TQA*	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)*	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle*	-	-	-	-	-	-	-	
Fourier (Aer)	81.1 ± 1.2	75.7 ± 1.5	71.5 ± 2.1	76.3 ± 1.4	10 [40]	10 [39]	10 [40]	10
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Interp (Aer)	81.1 ± 1.2	75.7 ± 1.5	71.5 ± 2.1	76.3 ± 1.4	10 [40]	10 [39]	10 [40]	10
Linear Ramp (Aer)*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)*	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
PP								
Fourier	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle*	-	-	-	-	-	-	-	
Interp	81.5 ± 1.2	75.4 ± 1.2	72.7 ± 1.7	76.8 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	74.4	-	75.2 ± 0.3	-	1 [106]	-	2 [40
TQA	-	-	-	-	-	-	-	
TQA*	-	-	-	-	-	-	-	
SV								
Fourier	78.8 ± 3.1	79.3 ± 3.0	74.8 ± 4.2	82.0 ± 4.1	61 [10–20]	10 [12]	699 [10–20]	570 [
Fixed Angle	78.4 ± 2.6	-	-	80.7 ± 3.1	55 [10–20]	-	-	289 [
Fixed Angle*	79.1 ± 2.9	-	-	81.7 ± 4.3	55 [10–20]	-	-	273 [
Interp	78.8 ± 2.9	79.3 ± 3.0	74.6 ± 3.9	81.9 ± 4.2	83 [10–20]	10 [12]	805 [10–22]	382 [
Linear Ramp*	-	-	-	-	-	-	-	
TQA	77.6 ± 2.6	-	71.0 ± 5.1	79.0 ± 2.9	65 [10–20]	-	592 [10–17]	280 [
TQA*	78.8 ± 3.0	-	74.7 ± 3.6	78.1 ± 2.1	63 [10–20]	-	584 [10–17]	60 [1
Recursive TS	78.8 ± 2.9	79.3 ± 3.0	75.0 ± 4.8	82.0 ± 4.3	81 [10–20]	10 [12]	481 [10–22]	368 [

Table 3: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

2 Tables for depth $P = 2$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle (Aer)	-	-	-	10 [40]
Fixed Angle (Aer)*	-	-	-	10 [40]
Fixed Angle	10 [50]	-	90 [100]	40 [30–70]
Fixed Angle*	10 [50]	-	90 [100]	40 [30–70]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)*	-	-	10 [100]	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
TQA (Aer)*	10 [40]	10 [39]	10 [40]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
TQA*	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Fixed Angle	10 [50]	-	90 [100]	40 [30–70]
Fixed Angle*	10 [50]	-	90 [100]	40 [30–70]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	51 [30–100]
Linear Ramp*	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
TQA*	20 [40–50]	10 [39]	130 [40–100]	50 [30–70]
		SV		
Fourier	61 [10–20]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	72 [10–20]	-	294 [10–22]	372 [10–20]
Fixed Angle*	72 [10–20]	-	293 [10–22]	360 [10–20]
Interp	83 [10–20]	10 [12]	805 [10–22]	372 [10–20]
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	804 [10–22]	361 [10–20]
TQA*	83 [10–20]	10 [12]	803 [10–22]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	415 [10–22]	353 [10–20]

Table 4: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4
Fixed Angle (Aer)	-	-	-	78.3 ± 2.8
Fixed Angle (Aer)*	-	-	-	79.0 ± 2.6
Fixed Angle	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4
Fourier (Aer)	70.5 ± 2.6	75.7 ± 4.4	70.9 ± 1.8	75.4 ± 2.2
Interp	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0
Interp (Aer)	77.5 ± 1.8	82.1 ± 0.9	77.6 ± 2.1	71.1 ± 2.5
Linear Ramp (Aer)*	-	-	68.4 ± 0.5	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.1 ± 1.7	73.7 ± 1.9
TQA (Aer)*	75.8 ± 1.3	80.7 ± 2.2	77.1 ± 2.3	77.3 ± 3.4
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0
PP				
Fourier	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5
Fixed Angle	86.9 ± 1.0	-	79.4 ± 1.8	83.6 ± 1.4
Fixed Angle*	87.8 ± 1.0	-	79.4 ± 1.8	83.6 ± 1.4
Interp	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4
Linear Ramp*	-	-	81.6 ± 1.1	-
Recursive TS	-	-	-	-
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4
SV				
Fourier	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9
Fixed Angle	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3
Interp	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6
Linear Ramp*	-	-	-	-
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3

Table 5: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
				MPS				
Fourier	70.6 ± 1.8	72.1 ± 8.3	76.1 ± 4.3	75.2 ± 4.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle (Aer)	-	-	-	78.3 ± 2.8	-	-	-	10
Fixed Angle (Aer)*	-	-	-	79.0 ± 2.6	-	-	-	10
Fixed Angle	77.2 ± 0.9	-	79.6 ± 2.6	76.6 ± 4.8	10 [50]	-	90 [100]	40 [3
Fixed Angle*	80.4 ± 1.8	-	79.9 ± 2.4	77.3 ± 4.4	10 [50]	-	90 [100]	40 [3
Fourier (Aer)	70.5 ± 2.6	75.7 ± 4.4	70.9 ± 1.8	75.4 ± 2.2	10 [40]	10 [39]	10 [40]	10
Interp	79.7 ± 2.0	75.9 ± 6.2	79.2 ± 3.0	75.6 ± 4.0	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Interp (Aer)	77.5 ± 1.8	82.1 ± 0.9	77.6 ± 2.1	71.1 ± 2.5	10 [40]	10 [39]	10 [40]	10
Linear Ramp (Aer)*	-	-	68.4 ± 0.5	-	-	-	10 [100]	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	69.2 ± 2.0	73.1 ± 1.6	67.1 ± 1.7	73.7 ± 1.9	10 [40]	10 [39]	10 [40]	10
TQA (Aer)*	75.8 ± 1.3	80.7 ± 2.2	77.1 ± 2.3	77.3 ± 3.4	10 [40]	10 [39]	10 [40]	10
TQA	69.3 ± 1.1	73.0 ± 1.7	70.0 ± 3.9	72.8 ± 2.5	20 [40–50]	10 [39]	130 [40–100]	50 [3
TQA*	77.4 ± 1.3	83.1 ± 1.6	79.2 ± 3.0	77.6 ± 4.0	20 [40–50]	10 [39]	130 [40–100]	50 [3
				PP				
Fourier	86.7 ± 2.1	80.8 ± 2.2	79.1 ± 1.7	80.2 ± 2.5	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Fixed Angle	86.9 ± 1.0	-	79.4 ± 1.8	83.6 ± 1.4	10 [50]	-	90 [100]	40 [3
Fixed Angle*	87.8 ± 1.0	-	79.4 ± 1.8	83.6 ± 1.4	10 [50]	-	90 [100]	40 [3
Interp	87.4 ± 1.2	82.7 ± 1.2	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	31 [39–144]	130 [40–100]	51 [3
Linear Ramp*	-	-	81.6 ± 1.1	-	-	-	10 [100]	
Recursive TS	-	-	-	-	-	-	-	
TQA	78.9 ± 1.3	73.0 ± 1.7	70.6 ± 3.0	76.9 ± 1.9	20 [40–50]	10 [39]	130 [40–100]	50 [3
TQA*	87.4 ± 1.2	83.1 ± 1.6	79.6 ± 2.0	83.8 ± 1.4	20 [40–50]	10 [39]	130 [40–100]	50 [3
				SV				
Fourier	75.8 ± 9.1	77.0 ± 3.9	76.4 ± 5.8	80.6 ± 8.9	61 [10–20]	10 [12]	699 [10–20]	567 [
Fixed Angle	84.1 ± 2.1	-	82.2 ± 2.9	87.2 ± 3.0	72 [10–20]	-	294 [10–22]	372 [
Fixed Angle*	85.5 ± 2.3	-	82.6 ± 2.9	88.1 ± 3.3	72 [10–20]	-	293 [10–22]	360 [
Interp	85.3 ± 2.4	85.5 ± 1.8	82.5 ± 4.1	86.5 ± 4.6	83 [10–20]	10 [12]	805 [10–22]	372 [
Linear Ramp*	-	-	-	-	-	-	-	
TQA	76.0 ± 3.2	73.0 ± 5.0	72.4 ± 4.2	75.7 ± 4.4	83 [10–20]	10 [12]	804 [10–22]	361 [
TQA*	85.5 ± 2.4	86.6 ± 2.5	82.8 ± 3.1	87.3 ± 3.3	83 [10–20]	10 [12]	803 [10–22]	136 [
Recursive TS	85.2 ± 2.4	87.1 ± 1.9	81.3 ± 2.8	86.1 ± 4.3	81 [10–20]	10 [12]	415 [10–22]	353 [

Table 6: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

3 Tables for depth $P = 3$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	30 [30–70]
Fixed Angle*	-	-	-	30 [30–70]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA*	-	-	10 [40]	30 [30–70]
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Fixed Angle	-	-	-	30 [30–70]
Fixed Angle*	-	-	-	30 [30–70]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	41 [30–100]
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	10 [40]	30 [30–70]
TQA*	-	-	10 [40]	30 [30–70]
SV				
Fourier	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	72 [10–20]	-	266 [10–20]	372 [10–20]
Fixed Angle*	72 [10–20]	-	266 [10–20]	364 [10–20]
Interp	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	803 [10–22]	357 [10–20]
TQA*	82 [10–20]	10 [12]	803 [10–22]	135 [10–20]
Recursive TS	81 [10–20]	10 [12]	415 [10–22]	357 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	81.6 ± 6.2
Fixed Angle*	-	-	-	83.3 ± 5.2
Fourier (Aer)	80.0 ± 4.1	79.6 ± 2.4	75.9 ± 8.1	77.1 ± 2.5
Interp	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7
Interp (Aer)	83.4 ± 3.0	86.1 ± 1.1	87.9 ± 1.6	73.4 ± 3.8
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9
TQA*	-	-	88.8 ± 1.1	81.5 ± 6.3
PP				
Fourier	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3
Fixed Angle	-	-	-	87.3 ± 1.3
Fixed Angle*	-	-	-	87.3 ± 1.3
Interp	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4
TQA*	-	-	79.7 ± 2.4	87.3 ± 1.3
SV				
Fourier	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1
Fixed Angle	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2
Fixed Angle*	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7
Interp	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1
Linear Ramp*	-	-	-	-
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5
TQA*	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8

Table 8: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	84.2 ± 6.6	75.6 ± 6.8	80.1 ± 4.4	78.0 ± 5.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)⋆	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	81.6 ± 6.2	-	-	-	30 [3
Fixed Angle⋆	-	-	-	83.3 ± 5.2	-	-	-	30 [3
Fourier (Aer)	80.0 ± 4.1	79.6 ± 2.4	75.9 ± 8.1	77.1 ± 2.5	10 [40]	10 [39]	10 [40]	10
Interp	87.9 ± 1.5	79.0 ± 7.0	85.7 ± 1.6	81.4 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Interp (Aer)	83.4 ± 3.0	86.1 ± 1.1	87.9 ± 1.6	73.4 ± 3.8	10 [40]	10 [39]	10 [40]	10
Linear Ramp (Aer)⋆	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)⋆	-	-	-	-	-	-	-	
TQA	-	-	82.1 ± 1.8	76.2 ± 3.9	-	-	10 [40]	30 [3
TQA⋆	-	-	88.8 ± 1.1	81.5 ± 6.3	-	-	10 [40]	30 [3
PP								
Fourier	86.7 ± 1.9	83.9 ± 2.7	81.0 ± 2.6	84.6 ± 2.3	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Fixed Angle	-	-	-	87.3 ± 1.3	-	-	-	30 [3
Fixed Angle⋆	-	-	-	87.3 ± 1.3	-	-	-	30 [3
Interp	89.0 ± 1.7	86.7 ± 1.1	83.3 ± 1.9	87.3 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	41 [3
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA	-	-	68.0 ± 1.6	80.7 ± 1.4	-	-	10 [40]	30 [3
TQA⋆	-	-	79.7 ± 2.4	87.3 ± 1.3	-	-	10 [40]	30 [3
SV								
Fourier	83.2 ± 5.9	79.2 ± 4.5	78.9 ± 7.6	88.4 ± 4.1	59 [10–19]	10 [12]	699 [10–20]	567 [
Fixed Angle	87.1 ± 1.8	-	87.2 ± 2.5	89.9 ± 2.2	72 [10–20]	-	266 [10–20]	372 [
Fixed Angle⋆	89.1 ± 1.9	-	87.8 ± 2.5	91.6 ± 2.7	72 [10–20]	-	266 [10–20]	364 [
Interp	88.8 ± 2.0	89.8 ± 1.5	87.4 ± 3.8	89.7 ± 4.1	82 [10–20]	10 [12]	803 [10–22]	372 [
Linear Ramp⋆	-	-	-	-	-	-	-	
TQA	81.1 ± 4.0	79.7 ± 4.8	77.3 ± 5.2	80.1 ± 5.5	83 [10–20]	10 [12]	803 [10–22]	357 [
TQA⋆	87.6 ± 2.2	88.2 ± 2.0	85.1 ± 3.8	89.9 ± 2.6	82 [10–20]	10 [12]	803 [10–22]	135 [
Recursive TS	88.0 ± 2.5	90.1 ± 1.8	84.9 ± 2.4	88.5 ± 3.8	81 [10–20]	10 [12]	415 [10–22]	357 [

Table 9: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

4 Tables for depth $P = 4$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	10 [40]
Fixed Angle (Aer)*	-	-	-	10 [40]
Fixed Angle	-	-	30 [100]	10 [40]
Fixed Angle*	-	-	30 [100]	10 [40]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
TQA (Aer)*	10 [40]	10 [39]	10 [40]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	-	30 [100]	10 [40]
Fixed Angle*	-	-	30 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA*	20 [40–50]	10 [39]	130 [40–100]	20 [40]
SV				
Fourier	59 [10–19]	10 [12]	699 [10–20]	567 [10–20]
Fixed Angle	52 [10–20]	-	95 [10–20]	158 [10–20]
Fixed Angle*	52 [10–20]	-	95 [10–20]	155 [10–20]
Interp	82 [10–20]	10 [12]	803 [10–22]	372 [10–20]
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	803 [10–22]	354 [10–20]
TQA*	82 [10–20]	10 [12]	803 [10–22]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	414 [10–22]	352 [10–20]

Table 10: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0
Fixed Angle (Aer)	-	-	-	91.2 ± 1.7
Fixed Angle (Aer)*	-	-	-	93.1 ± 1.0
Fixed Angle	-	-	90.4 ± 1.5	90.4 ± 1.2
Fixed Angle*	-	-	90.5 ± 1.4	92.6 ± 0.8
Fourier (Aer)	74.3 ± 4.1	82.4 ± 2.6	84.8 ± 4.0	79.7 ± 3.3
Interp	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0
Interp (Aer)	87.8 ± 2.2	88.9 ± 1.1	92.4 ± 1.1	80.4 ± 6.6
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	85.7 ± 2.9	78.5 ± 2.8
TQA (Aer)*	80.7 ± 4.4	84.8 ± 0.9	90.6 ± 1.7	82.5 ± 3.4
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8
TQA*	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4
PP				
Fourier	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9
Fixed Angle	-	-	84.6 ± 1.5	89.9 ± 1.6
Fixed Angle*	-	-	86.5 ± 1.7	89.9 ± 1.6
Interp	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8
TQA*	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5
SV				
Fourier	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4
Fixed Angle	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2
Fixed Angle*	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2
Interp	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8
Linear Ramp*	-	-	-	-
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6
TQA*	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5

Table 11: **Avg. Approx. Ratio ± standard deviation in percentage for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	R
MPS								
Fourier	77.4 ± 6.4	76.0 ± 8.2	83.0 ± 4.4	88.0 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Fixed Angle (Aer)	-	-	-	91.2 ± 1.7	-	-	-	10
Fixed Angle (Aer)*	-	-	-	93.1 ± 1.0	-	-	-	10
Fixed Angle	-	-	90.4 ± 1.5	90.4 ± 1.2	-	-	30 [100]	10
Fixed Angle*	-	-	90.5 ± 1.4	92.6 ± 0.8	-	-	30 [100]	10
Fourier (Aer)	74.3 ± 4.1	82.4 ± 2.6	84.8 ± 4.0	79.7 ± 3.3	10 [40]	10 [39]	10 [40]	10
Interp	92.1 ± 1.1	79.5 ± 8.6	89.9 ± 1.1	90.5 ± 6.0	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Interp (Aer)	87.8 ± 2.2	88.9 ± 1.1	92.4 ± 1.1	80.4 ± 6.6	10 [40]	10 [39]	10 [40]	10
Linear Ramp (Aer)*	-	-	-	-	-	-	-	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	68.9 ± 1.9	81.5 ± 1.0	85.7 ± 2.9	78.5 ± 2.8	10 [40]	10 [39]	10 [40]	10
TQA (Aer)*	80.7 ± 4.4	84.8 ± 0.9	90.6 ± 1.7	82.5 ± 3.4	10 [40]	10 [39]	10 [40]	10
TQA	70.5 ± 3.4	82.0 ± 1.6	84.2 ± 2.4	80.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20
TQA*	79.0 ± 7.0	86.9 ± 1.1	88.5 ± 1.9	88.1 ± 3.4	20 [40–50]	10 [39]	130 [40–100]	20
PP								
Fourier	88.0 ± 2.1	86.4 ± 2.7	82.4 ± 2.9	87.4 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Fixed Angle	-	-	84.6 ± 1.5	89.9 ± 1.6	-	-	30 [100]	10
Fixed Angle*	-	-	86.5 ± 1.7	89.9 ± 1.6	-	-	30 [100]	10
Interp	90.7 ± 1.9	89.3 ± 1.1	85.3 ± 1.8	89.3 ± 1.6	20 [40–50]	31 [39–144]	130 [40–100]	21 [
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA	83.7 ± 1.4	82.0 ± 1.6	75.7 ± 4.8	85.4 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	20
TQA*	89.2 ± 1.9	87.3 ± 1.3	83.5 ± 1.9	88.0 ± 1.5	20 [40–50]	10 [39]	130 [40–100]	20
SV								
Fourier	78.3 ± 10.6	83.6 ± 3.3	80.9 ± 8.0	90.3 ± 4.4	59 [10–19]	10 [12]	699 [10–20]	567
Fixed Angle	89.0 ± 2.2	-	91.0 ± 2.3	92.8 ± 2.2	52 [10–20]	-	95 [10–20]	158
Fixed Angle*	91.5 ± 1.7	-	91.9 ± 2.3	93.4 ± 2.2	52 [10–20]	-	95 [10–20]	155
Interp	91.3 ± 1.7	91.8 ± 1.5	90.4 ± 3.8	91.0 ± 4.8	82 [10–20]	10 [12]	803 [10–22]	372
Linear Ramp*	-	-	-	-	-	-	-	
TQA	84.6 ± 3.9	82.0 ± 4.9	79.8 ± 6.0	84.0 ± 5.6	83 [10–20]	10 [12]	803 [10–22]	354
TQA*	89.4 ± 1.9	89.5 ± 1.9	87.8 ± 2.7	91.1 ± 2.5	82 [10–20]	10 [12]	803 [10–22]	136
Recursive TS	90.3 ± 2.2	91.4 ± 1.7	87.6 ± 2.3	90.1 ± 3.5	81 [10–20]	10 [12]	414 [10–22]	352

Table 12: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

5 Tables for depth $P = 5$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	10 [40]
Fixed Angle*	-	-	-	10 [40]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)*	-	10 [144]	10 [100]	-
Linear Ramp*	-	-	-	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA*	-	20 [105–144]	10 [40]	10 [40]
		PP		
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	-	-	10 [40]
Fixed Angle*	-	-	-	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp*	-	10 [144]	10 [100]	-
Recursive TS	-	1 [106]	-	2 [40–100]
TQA	-	20 [105–144]	10 [40]	10 [40]
TQA*	-	20 [105–144]	10 [40]	10 [40]
		SV		
Fourier	58 [10–19]	10 [12]	699 [10–20]	266 [10–20]
Fixed Angle	39 [10–20]	-	50 [10–19]	121 [10–20]
Fixed Angle*	39 [10–20]	-	49 [10–19]	115 [10–20]
Interp	82 [10–20]	10 [12]	693 [10–20]	372 [10–20]
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	359 [10–20]
TQA*	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	81 [10–20]	10 [12]	276 [10–20]	348 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
		MPS		
Fourier	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	93.8 ± 0.6
Fixed Angle*	-	-	-	95.0 ± 0.5
Fourier (Aer)	82.1 ± 6.3	84.7 ± 2.1	90.9 ± 1.2	80.5 ± 4.8
Interp	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6
Interp (Aer)	91.6 ± 1.6	90.8 ± 1.4	94.6 ± 0.8	86.3 ± 4.7
Linear Ramp (Aer)*	-	85.7 ± 1.0	64.5 ± 2.1	-
Linear Ramp*	-	-	-	-
Recursive TS	-	71.1	-	83.7 ± 7.3
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5
TQA*	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4
		PP		
Fourier	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5
Fixed Angle	-	-	-	91.5 ± 1.5
Fixed Angle*	-	-	-	91.7 ± 1.5
Interp	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8
Linear Ramp*	-	88.8 ± 0.8	86.9 ± 1.3	-
Recursive TS	-	88.2	-	89.0 ± 2.3
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4
TQA*	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5
		SV		
Fourier	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7
Fixed Angle	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0
Fixed Angle*	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0
Interp	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4
Linear Ramp*	-	-	-	-
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3
TQA*	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fourier	82.0 ± 7.5	77.1 ± 8.9	85.8 ± 4.6	90.0 ± 8.3	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)*	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	93.8 ± 0.6	-	-	-	
Fixed Angle*	-	-	-	95.0 ± 0.5	-	-	-	
Fourier (Aer)	82.1 ± 6.3	84.7 ± 2.1	90.9 ± 1.2	80.5 ± 4.8	10 [40]	10 [39]	10 [40]	
Interp	94.5 ± 1.2	81.5 ± 8.7	92.4 ± 1.1	93.4 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp (Aer)	91.6 ± 1.6	90.8 ± 1.4	94.6 ± 0.8	86.3 ± 4.7	10 [40]	10 [39]	10 [40]	
Linear Ramp (Aer)*	-	85.7 ± 1.0	64.5 ± 2.1	-	-	10 [144]	10 [100]	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	71.1	-	83.7 ± 7.3	-	1 [106]	-	2
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)*	-	-	-	-	-	-	-	
TQA	-	73.4 ± 4.6	88.2 ± 1.2	82.6 ± 1.5	-	20 [105–144]	10 [40]	
TQA*	-	75.2 ± 4.8	91.4 ± 1.0	90.2 ± 1.4	-	20 [105–144]	10 [40]	
PP								
Fourier	89.6 ± 1.9	88.4 ± 2.7	83.5 ± 2.5	89.0 ± 1.5	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle	-	-	-	91.5 ± 1.5	-	-	-	
Fixed Angle*	-	-	-	91.7 ± 1.5	-	-	-	
Interp	91.9 ± 2.0	91.1 ± 1.0	86.6 ± 1.8	90.5 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	2
Linear Ramp*	-	88.8 ± 0.8	86.9 ± 1.3	-	-	10 [144]	10 [100]	
Recursive TS	-	88.2	-	89.0 ± 2.3	-	1 [106]	-	2
TQA	-	83.8 ± 1.1	68.6 ± 1.7	85.9 ± 1.4	-	20 [105–144]	10 [40]	
TQA*	-	88.2 ± 0.8	83.5 ± 2.0	89.7 ± 1.5	-	20 [105–144]	10 [40]	
SV								
Fourier	80.0 ± 10.3	79.2 ± 6.8	81.4 ± 8.3	85.2 ± 10.7	58 [10–19]	10 [12]	699 [10–20]	20
Fixed Angle	90.0 ± 2.3	-	94.0 ± 1.8	94.9 ± 2.0	39 [10–20]	-	50 [10–19]	1
Fixed Angle*	93.2 ± 1.5	-	95.2 ± 1.7	95.3 ± 2.0	39 [10–20]	-	49 [10–19]	1
Interp	93.0 ± 1.5	92.5 ± 0.9	92.6 ± 4.1	92.5 ± 4.4	82 [10–20]	10 [12]	693 [10–20]	3
Linear Ramp*	-	-	-	-	-	-	-	
TQA	86.7 ± 3.7	83.7 ± 4.9	81.9 ± 6.4	86.6 ± 5.3	83 [10–20]	10 [12]	659 [10–20]	3
TQA*	91.6 ± 1.5	90.7 ± 2.0	90.6 ± 3.4	93.1 ± 2.2	81 [10–20]	10 [12]	653 [10–20]	1
Recursive TS	91.9 ± 2.1	92.6 ± 1.3	89.8 ± 3.5	91.1 ± 3.5	81 [10–20]	10 [12]	276 [10–20]	3

Table 15: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

6 Tables for depth $P = 6$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	10 [40]
Fixed Angle (Aer)⋆	-	-	-	10 [40]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)⋆	-	-	10 [100]	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	10 [40]	10 [40]
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
PP				
Fourier	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	130 [40–100]	21 [40–100]
Linear Ramp⋆	-	-	10 [100]	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	130 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	130 [40–100]	20 [40]
SV				
Fourier	56 [10–19]	10 [12]	693 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	19 [10–18]	59 [10–20]
Fixed Angle⋆	16 [11–17]	-	19 [10–18]	59 [10–20]
Interp	82 [10–20]	10 [12]	686 [10–20]	372 [10–20]
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA⋆	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	80 [10–20]	10 [12]	269 [10–20]	348 [10–20]

Table 16: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7
Fixed Angle (Aer)	-	-	-	96.6 ± 0.5
Fixed Angle (Aer)*	-	-	-	97.2 ± 0.2
Fixed Angle	-	-	95.1 ± 1.0	96.1 ± 0.5
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6
Fourier (Aer)	79.3 ± 7.1	85.1 ± 2.5	90.3 ± 2.8	86.1 ± 3.2
Interp	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6
Interp (Aer)	92.8 ± 1.3	92.4 ± 1.2	95.9 ± 0.6	89.3 ± 3.4
Linear Ramp (Aer)*	-	-	65.4 ± 3.0	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	88.6 ± 2.4	86.3 ± 2.3
TQA (Aer)*	83.3 ± 3.8	86.8 ± 1.3	92.4 ± 1.7	93.2 ± 1.1
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3
PP				
Fourier	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8
Fixed Angle	-	-	85.5 ± 1.6	92.2 ± 1.6
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5
Interp	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9
Linear Ramp*	-	-	87.9 ± 1.3	-
Recursive TS	-	-	-	-
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6
SV				
Fourier	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8
Fixed Angle	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4
Interp	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9
Linear Ramp*	-	-	-	-
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4

Table 17: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
				MPS				
Fourier	78.9 ± 7.6	76.6 ± 10.0	87.9 ± 3.5	92.2 ± 6.7	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle (Aer)	-	-	-	96.6 ± 0.5	-	-	-	
Fixed Angle (Aer)*	-	-	-	97.2 ± 0.2	-	-	-	
Fixed Angle	-	-	95.1 ± 1.0	96.1 ± 0.5	-	-	10 [100]	
Fixed Angle*	-	-	95.1 ± 1.1	96.9 ± 0.6	-	-	10 [100]	
Fourier (Aer)	79.3 ± 7.1	85.1 ± 2.5	90.3 ± 2.8	86.1 ± 3.2	10 [40]	10 [39]	10 [40]	
Interp	95.8 ± 1.1	81.9 ± 9.7	93.9 ± 1.0	94.3 ± 3.6	20 [40–50]	31 [39–144]	130 [40–100]	2
Interp (Aer)	92.8 ± 1.3	92.4 ± 1.2	95.9 ± 0.6	89.3 ± 3.4	10 [40]	10 [39]	10 [40]	
Linear Ramp (Aer)*	-	-	65.4 ± 3.0	-	-	-	10 [100]	
Linear Ramp*	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	74.2 ± 2.8	84.8 ± 0.9	88.6 ± 2.4	86.3 ± 2.3	10 [40]	10 [39]	10 [40]	
TQA (Aer)*	83.3 ± 3.8	86.8 ± 1.3	92.4 ± 1.7	93.2 ± 1.1	10 [40]	10 [39]	10 [40]	
TQA	71.7 ± 6.0	85.6 ± 1.4	88.0 ± 2.1	89.9 ± 2.7	20 [40–50]	10 [39]	130 [40–100]	
TQA*	85.5 ± 6.7	88.8 ± 0.9	91.1 ± 1.4	93.9 ± 1.3	20 [40–50]	10 [39]	130 [40–100]	
				PP				
Fourier	90.7 ± 1.9	89.3 ± 2.8	84.8 ± 2.2	89.7 ± 1.8	20 [40–50]	31 [39–144]	130 [40–100]	2
Fixed Angle	-	-	85.5 ± 1.6	92.2 ± 1.6	-	-	10 [100]	
Fixed Angle*	-	-	89.1 ± 1.8	92.5 ± 1.5	-	-	10 [100]	
Interp	92.7 ± 2.1	92.2 ± 0.9	87.6 ± 1.8	91.6 ± 1.9	20 [40–50]	31 [39–144]	130 [40–100]	2
Linear Ramp*	-	-	87.9 ± 1.3	-	-	-	10 [100]	
Recursive TS	-	-	-	-	-	-	-	
TQA	84.6 ± 1.9	85.6 ± 1.4	76.5 ± 5.9	88.0 ± 1.8	20 [40–50]	10 [39]	130 [40–100]	
TQA*	91.3 ± 2.4	89.8 ± 1.0	85.2 ± 2.2	90.7 ± 1.6	20 [40–50]	10 [39]	130 [40–100]	
				SV				
Fourier	79.8 ± 11.1	81.3 ± 11.1	81.1 ± 9.4	84.7 ± 11.8	56 [10–19]	10 [12]	693 [10–20]	2
Fixed Angle	90.5 ± 2.5	-	95.7 ± 1.3	95.9 ± 1.3	16 [11–17]	-	19 [10–18]	5
Fixed Angle*	94.5 ± 1.5	-	96.6 ± 1.4	96.2 ± 1.4	16 [11–17]	-	19 [10–18]	5
Interp	94.2 ± 1.3	93.6 ± 1.4	93.8 ± 4.0	93.1 ± 4.9	82 [10–20]	10 [12]	686 [10–20]	3
Linear Ramp*	-	-	-	-	-	-	-	
TQA	88.1 ± 3.5	85.6 ± 4.4	82.4 ± 7.3	88.4 ± 5.0	83 [10–20]	10 [12]	659 [10–20]	3
TQA*	93.0 ± 1.4	91.2 ± 2.2	91.9 ± 3.1	94.2 ± 2.2	81 [10–20]	10 [12]	653 [10–20]	1
Recursive TS	93.1 ± 1.9	93.1 ± 1.3	91.0 ± 2.7	92.2 ± 3.4	80 [10–20]	10 [12]	269 [10–20]	3

Table 18: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

7 Tables for depth $P = 7$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)⋆	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle⋆	-	-	-	-
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)⋆	-	-	-	-
TQA	-	-	-	-
TQA⋆	-	-	-	-
		PP		
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle	-	-	-	-
Fixed Angle⋆	-	-	-	-
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA⋆	-	-	-	-
		SV		
Fourier	54 [10–18]	10 [12]	688 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	60 [10–20]
Fixed Angle⋆	16 [11–17]	-	18 [10–18]	59 [10–20]
Interp	82 [10–20]	10 [12]	681 [10–20]	372 [10–20]
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	360 [10–20]
TQA⋆	81 [10–20]	10 [12]	653 [10–20]	135 [10–20]
Recursive TS	80 [10–20]	10 [12]	270 [10–20]	349 [10–20]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	81.3 ± 6.9	86.4 ± 3.1	90.1 ± 3.7	88.2 ± 6.1
Interp	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9
Interp (Aer)	94.0 ± 1.5	93.6 ± 1.0	96.7 ± 0.6	90.4 ± 2.4
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2
Fixed Angle	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2
Fixed Angle*	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2
Interp	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4
Linear Ramp*	-	-	-	-
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8
TQA*	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4

Table 20: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances			
	ER	HH	L2F	RR	ER	HH	L2F	
MPS								
Fourier	81.1 ± 7.1	77.6 ± 10.1	88.1 ± 4.1	91.0 ± 7.6	20 [40–50]	31 [39–144]	120 [40–100]	2
Fixed Angle (Aer)	-	-	-	-	-	-	-	
Fixed Angle (Aer)⋆	-	-	-	-	-	-	-	
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle⋆	-	-	-	-	-	-	-	
Fourier (Aer)	81.3 ± 6.9	86.4 ± 3.1	90.1 ± 3.7	88.2 ± 6.1	10 [40]	10 [39]	10 [40]	
Interp	96.4 ± 1.3	83.7 ± 9.4	94.9 ± 1.0	95.1 ± 3.9	20 [40–50]	31 [39–144]	120 [40–100]	2
Interp (Aer)	94.0 ± 1.5	93.6 ± 1.0	96.7 ± 0.6	90.4 ± 2.4	10 [40]	10 [39]	10 [40]	
Linear Ramp (Aer)⋆	-	-	-	-	-	-	-	
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA (Aer)	-	-	-	-	-	-	-	
TQA (Aer)⋆	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
PP								
Fourier	90.0 ± 2.8	89.9 ± 2.7	85.2 ± 2.1	90.1 ± 1.8	20 [40–50]	31 [39–144]	120 [40–100]	2
Fixed Angle	-	-	-	-	-	-	-	
Fixed Angle⋆	-	-	-	-	-	-	-	
Interp	93.3 ± 2.1	93.1 ± 0.9	88.3 ± 1.8	91.5 ± 2.0	20 [40–50]	31 [39–144]	120 [40–100]	2
Linear Ramp⋆	-	-	-	-	-	-	-	
Recursive TS	-	-	-	-	-	-	-	
TQA	-	-	-	-	-	-	-	
TQA⋆	-	-	-	-	-	-	-	
SV								
Fourier	80.0 ± 10.9	81.7 ± 10.2	81.0 ± 9.8	84.7 ± 12.2	54 [10–18]	10 [12]	688 [10–20]	2
Fixed Angle	91.0 ± 3.0	-	96.4 ± 1.0	96.8 ± 1.2	16 [11–17]	-	18 [10–18]	6
Fixed Angle⋆	95.5 ± 1.6	-	97.5 ± 1.1	97.2 ± 1.2	16 [11–17]	-	18 [10–18]	5
Interp	95.1 ± 1.3	94.3 ± 1.2	94.7 ± 4.2	93.3 ± 5.4	82 [10–20]	10 [12]	681 [10–20]	3
Linear Ramp⋆	-	-	-	-	-	-	-	
TQA	89.2 ± 3.4	87.0 ± 4.1	83.0 ± 7.8	89.3 ± 4.8	83 [10–20]	10 [12]	659 [10–20]	3
TQA⋆	93.9 ± 1.4	91.8 ± 2.2	92.5 ± 3.2	95.1 ± 1.8	81 [10–20]	10 [12]	653 [10–20]	1
Recursive TS	93.9 ± 1.9	93.8 ± 1.0	92.4 ± 4.1	92.9 ± 3.4	80 [10–20]	10 [12]	270 [10–20]	3

Table 21: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

8 Tables for depth $P = 8$

	ER	HH	L2F	RR
MPS				
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle (Aer)	-	-	-	10 [40]
Fixed Angle (Aer)⋆	-	-	-	10 [40]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Fourier (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Interp (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
Linear Ramp (Aer)⋆	-	-	-	-
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	10 [40]	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	10 [40]	10 [40]
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	120 [40–100]	20 [40]
PP				
Fourier	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Fixed Angle	-	-	10 [100]	10 [40]
Fixed Angle⋆	-	-	10 [100]	10 [40]
Interp	20 [40–50]	31 [39–144]	120 [40–100]	21 [40–100]
Linear Ramp⋆	-	-	-	-
Recursive TS	-	-	-	-
TQA	20 [40–50]	10 [39]	120 [40–100]	20 [40]
TQA⋆	20 [40–50]	10 [39]	120 [40–100]	20 [40]
SV				
Fourier	53 [10–18]	10 [12]	681 [10–20]	266 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle⋆	16 [11–17]	-	18 [10–18]	50 [10–20]
Interp	82 [10–20]	10 [12]	676 [10–20]	372 [10–20]
Linear Ramp⋆	-	-	-	-
TQA	83 [10–20]	10 [12]	659 [10–20]	355 [10–20]
TQA⋆	81 [10–20]	10 [12]	653 [10–20]	136 [10–20]
Recursive TS	80 [10–20]	10 [12]	265 [10–20]	348 [10–20]

Table 22: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3
Fixed Angle (Aer)	-	-	-	98.1 ± 0.3
Fixed Angle (Aer)*	-	-	-	98.4 ± 0.5
Fixed Angle	-	-	96.6 ± 0.8	97.9 ± 0.4
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3
Fourier (Aer)	79.5 ± 7.1	87.3 ± 3.2	90.4 ± 4.4	86.5 ± 8.7
Interp	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2
Interp (Aer)	94.3 ± 1.5	94.4 ± 1.1	97.2 ± 0.6	90.8 ± 2.7
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	86.8 ± 4.1	91.1 ± 1.4
TQA (Aer)*	86.3 ± 1.7	89.3 ± 1.0	93.3 ± 1.5	96.0 ± 0.5
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0
PP				
Fourier	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2
Fixed Angle	-	-	86.2 ± 1.7	92.8 ± 1.6
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6
Interp	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5
SV				
Fourier	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8
Fixed Angle	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1
Interp	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4
Linear Ramp*	-	-	-	-
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3

Table 23: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	81.3 ± 6.6	76.8 ± 10.2	88.2 ± 5.2	93.8 ± 6.3	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle (Aer)	-	-	-	98.1 ± 0.3	-	-	-
Fixed Angle (Aer)*	-	-	-	98.4 ± 0.5	-	-	-
Fixed Angle	-	-	96.6 ± 0.8	97.9 ± 0.4	-	-	10 [100]
Fixed Angle*	-	-	96.7 ± 0.8	98.3 ± 0.3	-	-	10 [100]
Fourier (Aer)	79.5 ± 7.1	87.3 ± 3.2	90.4 ± 4.4	86.5 ± 8.7	10 [40]	10 [39]	10 [40]
Interp	96.9 ± 1.3	84.3 ± 10.1	95.6 ± 1.0	96.1 ± 3.2	20 [40–50]	31 [39–144]	120 [40–100]
Interp (Aer)	94.3 ± 1.5	94.4 ± 1.1	97.2 ± 0.6	90.8 ± 2.7	10 [40]	10 [39]	10 [40]
Linear Ramp (Aer)*	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	79.5 ± 3.5	87.5 ± 0.8	86.8 ± 4.1	91.1 ± 1.4	10 [40]	10 [39]	10 [40]
TQA (Aer)*	86.3 ± 1.7	89.3 ± 1.0	93.3 ± 1.5	96.0 ± 0.5	10 [40]	10 [39]	10 [40]
TQA	82.2 ± 6.9	88.3 ± 1.2	89.3 ± 2.0	93.5 ± 2.0	20 [40–50]	10 [39]	120 [40–100]
TQA*	91.1 ± 3.7	90.3 ± 1.0	92.2 ± 1.2	96.4 ± 1.0	20 [40–50]	10 [39]	120 [40–100]
PP							
Fourier	91.0 ± 1.9	90.1 ± 3.0	85.7 ± 2.0	90.4 ± 2.2	20 [40–50]	31 [39–144]	120 [40–100]
Fixed Angle	-	-	86.2 ± 1.7	92.8 ± 1.6	-	-	10 [100]
Fixed Angle*	-	-	90.3 ± 1.9	93.4 ± 1.6	-	-	10 [100]
Interp	93.7 ± 2.0	93.9 ± 0.9	88.8 ± 1.8	91.9 ± 2.1	20 [40–50]	31 [39–144]	120 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	84.6 ± 2.2	88.2 ± 1.2	77.4 ± 6.7	89.9 ± 1.6	20 [40–50]	10 [39]	120 [40–100]
TQA*	91.7 ± 2.3	91.1 ± 0.7	85.6 ± 2.5	92.0 ± 1.5	20 [40–50]	10 [39]	120 [40–100]
SV							
Fourier	79.0 ± 12.3	79.4 ± 11.1	79.7 ± 11.2	84.1 ± 12.8	53 [10–18]	10 [12]	681 [10–20]
Fixed Angle	91.8 ± 3.2	-	96.8 ± 1.1	97.2 ± 1.1	16 [11–17]	-	18 [10–18]
Fixed Angle*	96.4 ± 1.4	-	98.1 ± 0.9	97.8 ± 1.1	16 [11–17]	-	18 [10–18]
Interp	95.8 ± 1.2	94.7 ± 1.1	95.3 ± 4.0	93.7 ± 5.4	82 [10–20]	10 [12]	676 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
TQA	90.2 ± 3.5	88.1 ± 4.0	83.5 ± 8.5	90.0 ± 4.9	83 [10–20]	10 [12]	659 [10–20]
TQA*	94.4 ± 1.3	92.3 ± 1.8	93.2 ± 3.3	95.7 ± 1.6	81 [10–20]	10 [12]	653 [10–20]
Recursive TS	94.6 ± 1.8	94.3 ± 1.0	93.3 ± 4.6	93.4 ± 3.3	80 [10–20]	10 [12]	265 [10–20]

Table 24: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

9 Tables for depth $P = 9$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	10 [40]	10 [39]	-	10 [40]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp (Aer)	10 [40]	10 [39]	-	10 [40]
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		PP		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
		SV		
Fourier	52 [10–18]	10 [12]	672 [10–20]	264 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	50 [10–20]
Fixed Angle*	16 [11–17]	-	18 [10–18]	50 [10–20]
Interp	82 [10–20]	10 [12]	673 [10–20]	371 [10–20]
Linear Ramp*	-	-	-	-
TQA	83 [10–20]	10 [12]	658 [10–20]	356 [10–20]
TQA*	81 [10–20]	10 [12]	652 [10–20]	135 [10–20]
Recursive TS	80 [10–20]	10 [12]	264 [10–20]	352 [10–20]

Table 25: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
MPS				
Fourier	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4
Fixed Angle (Aer)	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	88.7 ± 3.6	-	89.7 ± 3.2
Interp	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5
Interp (Aer)	94.1 ± 4.2	95.0 ± 1.0	-	93.5 ± 1.9
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7
Fixed Angle	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0
Fixed Angle*	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1
Interp	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1
Linear Ramp*	-	-	-	-
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9
TQA*	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2

Table 26: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio				Num. Instances		
	ER	HH	L2F	RR	ER	HH	L2F
MPS							
Fourier	82.0 ± 7.8	78.3 ± 9.7	88.6 ± 4.3	92.7 ± 7.4	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle (Aer)	-	-	-	-	-	-	-
Fixed Angle (Aer)*	-	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-	-
Fourier (Aer)	79.0 ± 6.7	88.7 ± 3.6	-	89.7 ± 3.2	10 [40]	10 [39]	-
Interp	97.2 ± 1.4	86.2 ± 9.3	96.2 ± 0.9	93.1 ± 4.5	20 [40–50]	31 [39–144]	111 [40–100]
Interp (Aer)	94.1 ± 4.2	95.0 ± 1.0	-	93.5 ± 1.9	10 [40]	10 [39]	-
Linear Ramp (Aer)*	-	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	-	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
PP							
Fourier	91.7 ± 1.6	90.6 ± 3.2	86.2 ± 1.9	91.0 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle	-	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-	-
Interp	94.1 ± 2.0	94.4 ± 0.8	89.2 ± 1.7	90.8 ± 2.3	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp*	-	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-	-
TQA	-	-	-	-	-	-	-
TQA*	-	-	-	-	-	-	-
SV							
Fourier	78.2 ± 12.7	78.0 ± 10.8	79.1 ± 11.6	83.5 ± 13.7	52 [10–18]	10 [12]	672 [10–20]
Fixed Angle	92.1 ± 3.4	-	97.0 ± 1.3	97.7 ± 1.0	16 [11–17]	-	18 [10–18]
Fixed Angle*	97.0 ± 1.4	-	98.5 ± 0.8	98.3 ± 1.1	16 [11–17]	-	18 [10–18]
Interp	96.3 ± 1.1	94.8 ± 1.2	95.7 ± 4.3	93.6 ± 6.1	82 [10–20]	10 [12]	673 [10–20]
Linear Ramp*	-	-	-	-	-	-	-
TQA	90.9 ± 3.7	89.1 ± 3.8	83.7 ± 9.2	90.8 ± 4.9	83 [10–20]	10 [12]	658 [10–20]
TQA*	95.0 ± 1.3	93.0 ± 1.6	93.6 ± 3.3	96.2 ± 1.7	81 [10–20]	10 [12]	652 [10–20]
Recursive TS	95.0 ± 1.9	94.7 ± 0.9	93.6 ± 5.2	93.6 ± 3.2	80 [10–20]	10 [12]	264 [10–20]

Table 27: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

10 Tables for depth $P = 10$

	ER	HH	L2F	RR
		MPS		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle (Aer)	-	-	-	10 [40]
Fixed Angle (Aer)⋆	-	-	-	10 [40]
Fixed Angle	-	-	10 [100]	11 [40–100]
Fixed Angle⋆	-	-	10 [100]	11 [40–100]
Fourier (Aer)	10 [40]	10 [39]	-	10 [40]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Interp (Aer)	10 [40]	10 [39]	-	10 [40]
Linear Ramp (Aer)⋆	10 [40]	30 [39–144]	44 [40–100]	10 [40]
Linear Ramp⋆	10 [40]	10 [39]	-	-
Recursive TS	-	-	-	-
TQA (Aer)	10 [40]	10 [39]	-	10 [40]
TQA (Aer)⋆	10 [40]	10 [39]	-	10 [40]
TQA	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
TQA⋆	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
		PP		
Fourier	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Fixed Angle	-	-	10 [100]	11 [40–100]
Fixed Angle⋆	-	-	10 [100]	11 [40–100]
Interp	20 [40–50]	31 [39–144]	111 [40–100]	11 [40–100]
Linear Ramp⋆	10 [40]	20 [39–144]	118 [40–100]	10 [40]
Recursive TS	-	-	-	-
TQA	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
TQA⋆	20 [40–50]	30 [39–144]	111 [40–100]	11 [40–100]
		SV		
Fourier	50 [10–17]	10 [12]	665 [10–20]	252 [10–20]
Fixed Angle	16 [11–17]	-	18 [10–18]	46 [10–24]
Fixed Angle⋆	16 [11–17]	-	18 [10–18]	43 [10–22]
Interp	82 [10–20]	10 [12]	664 [10–20]	353 [10–20]
Linear Ramp⋆	88 [10–20]	10 [12]	175 [10–20]	568 [10–20]
TQA	83 [10–20]	10 [12]	657 [10–20]	349 [10–22]
TQA⋆	81 [10–20]	9 [12]	651 [10–20]	147 [10–20]
Recursive TS	79 [10–20]	9 [12]	264 [10–20]	333 [10–20]

Table 28: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	L2F	RR
		MPS		
Fourier	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0
Fixed Angle (Aer)	-	-	-	98.8 ± 0.3
Fixed Angle (Aer)*	-	-	-	99.0 ± 0.4
Fixed Angle	-	-	97.4 ± 0.7	98.2 ± 1.4
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9
Fourier (Aer)	82.5 ± 7.8	87.7 ± 5.2	-	91.6 ± 3.7
Interp	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6
Interp (Aer)	94.3 ± 4.5	95.6 ± 0.8	-	92.0 ± 1.6
Linear Ramp (Aer)*	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-
Recursive TS	-	-	-	-
TQA (Aer)	85.9 ± 2.1	89.4 ± 0.9	-	94.1 ± 0.9
TQA (Aer)*	90.1 ± 1.9	90.6 ± 0.9	-	96.5 ± 0.6
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7
		PP		
Fourier	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6
Fixed Angle	-	-	86.7 ± 1.7	93.0 ± 1.6
Fixed Angle*	-	-	90.7 ± 1.9	93.6 ± 1.6
Interp	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6
Recursive TS	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6
		SV		
Fourier	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9
Fixed Angle	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9
Interp	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2

Table 29: **Avg. Approx. Ratio $\pm$ standard deviation in percentage for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors, with darker colors indicating better approximation ratios.

	Approximation Ratio ER	HH	L2F	RR	Num. Instances ER	HH	L2F
MPS							
Fourier	79.6 ± 6.1	77.4 ± 10.9	88.7 ± 5.6	92.8 ± 9.0	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle (Aer)	-	-	-	98.8 ± 0.3	-	-	-
Fixed Angle (Aer)*	-	-	-	99.0 ± 0.4	-	-	-
Fixed Angle	-	-	97.4 ± 0.7	98.2 ± 1.4	-	-	10 [100]
Fixed Angle*	-	-	97.6 ± 0.7	98.6 ± 0.9	-	-	10 [100]
Fourier (Aer)	82.5 ± 7.8	87.7 ± 5.2	-	91.6 ± 3.7	10 [40]	10 [39]	-
Interp	97.4 ± 1.5	86.6 ± 10.1	96.6 ± 0.9	93.8 ± 4.6	20 [40–50]	31 [39–144]	111 [40–100]
Interp (Aer)	94.3 ± 4.5	95.6 ± 0.8	-	92.0 ± 1.6	10 [40]	10 [39]	-
Linear Ramp (Aer)*	95.2 ± 0.9	91.9 ± 1.5	90.3 ± 4.8	96.1 ± 0.8	10 [40]	30 [39–144]	44 [40–100]
Linear Ramp*	96.8 ± 0.5	93.6 ± 0.7	-	-	10 [40]	10 [39]	-
Recursive TS	-	-	-	-	-	-	-
TQA (Aer)	85.9 ± 2.1	89.4 ± 0.9	-	94.1 ± 0.9	10 [40]	10 [39]	-
TQA (Aer)*	90.1 ± 1.9	90.6 ± 0.9	-	96.5 ± 0.6	10 [40]	10 [39]	-
TQA	87.7 ± 1.8	79.6 ± 9.2	90.0 ± 2.5	94.0 ± 3.3	20 [40–50]	31 [39–144]	111 [40–100]
TQA*	91.8 ± 2.9	80.8 ± 9.3	92.6 ± 1.7	96.5 ± 1.7	20 [40–50]	31 [39–144]	111 [40–100]
PP							
Fourier	91.5 ± 1.9	91.1 ± 3.2	86.0 ± 2.2	91.3 ± 2.6	20 [40–50]	31 [39–144]	111 [40–100]
Fixed Angle	-	-	86.7 ± 1.7	93.0 ± 1.6	-	-	10 [100]
Fixed Angle*	-	-	90.7 ± 1.9	93.6 ± 1.6	-	-	10 [100]
Interp	94.4 ± 1.9	94.9 ± 0.8	89.6 ± 1.7	90.1 ± 2.4	20 [40–50]	31 [39–144]	111 [40–100]
Linear Ramp*	94.8 ± 1.4	93.0 ± 0.8	87.6 ± 1.7	92.5 ± 1.6	10 [40]	20 [39–144]	118 [40–100]
Recursive TS	-	-	-	-	-	-	-
TQA	84.4 ± 2.4	90.0 ± 0.9	77.5 ± 7.8	90.8 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
TQA*	91.3 ± 2.5	91.8 ± 0.7	85.6 ± 3.0	92.3 ± 1.6	20 [40–50]	30 [39–144]	111 [40–100]
SV							
Fourier	77.5 ± 13.2	78.5 ± 11.7	77.9 ± 12.3	83.4 ± 13.9	50 [10–17]	10 [12]	665 [10–20]
Fixed Angle	92.4 ± 3.5	-	97.2 ± 1.3	98.1 ± 0.9	16 [11–17]	-	18 [10–18]
Fixed Angle*	97.4 ± 1.4	-	98.8 ± 0.6	98.7 ± 0.9	16 [11–17]	-	18 [10–18]
Interp	96.6 ± 1.1	94.8 ± 1.1	96.0 ± 3.9	94.0 ± 5.9	82 [10–20]	10 [12]	664 [10–20]
Linear Ramp*	95.0 ± 1.5	94.5 ± 1.4	93.5 ± 2.0	96.2 ± 1.8	88 [10–20]	10 [12]	175 [10–20]
TQA	91.3 ± 3.9	89.9 ± 3.6	83.8 ± 9.5	91.0 ± 5.1	83 [10–20]	10 [12]	657 [10–20]
TQA*	95.4 ± 1.4	93.2 ± 2.1	93.9 ± 3.2	96.7 ± 1.5	81 [10–20]	9 [12]	651 [10–20]
Recursive TS	95.3 ± 2.0	95.2 ± 0.8	93.4 ± 4.4	94.0 ± 3.2	79 [10–20]	9 [12]	264 [10–20]

Table 30: **Avg. Approx. Ratio and Number of Instances for Various Evaluation Methods for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Evaluation methods are shown in different colors for approximation ratios. Darker colors indicating better approximation ratios or more instances. Approximation ratios are in percentage, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

Summary Tables of MIS Data

11 Tables for depth $P = 1$

	ER	HH	RR
MPS			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	-	-	-
TQA*	-	-	-
PP			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fourier	32 [10–17]	-	284 [10–20]
Fixed Angle	37 [10–19]	-	38 [10–20]
Fixed Angle*	36 [10–18]	-	37 [10–20]
Interp	32 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-
TQA	37 [10–19]	-	280 [10–20]
TQA*	35 [10–18]	-	280 [10–20]
Recursive TS	-	-	-

Table 31: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
MPS			
Fourier	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Fourier (Aer)	-	-	-
Interp	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	-	-	-
TQA*	-	-	-
PP			
Fourier	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Interp	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fourier	123.6 ± 64.5	-	292.7 ± 140.4
Fixed Angle	69.7 ± 23.9	-	109.7 ± 36.3
Fixed Angle*	135.6 ± 76.5	-	245.2 ± 151.7
Interp	123.6 ± 64.5	-	292.7 ± 140.4
Linear Ramp*	-	-	-
TQA	135.0 ± 75.3	-	291.7 ± 140.8
TQA*	128.9 ± 66.1	-	293.4 ± 140.4
Recursive TS	-	-	-

Table 32: **Avg. Normalized Energy $\pm$ standard deviation for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Fourier (Aer)	-	-	-	-	-	-
Interp	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
PP						
Fourier	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Interp	407.0 ± 45.6	46.9 ± 0.0	92.5 ± 0.2	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fourier	123.6 ± 64.5	-	292.7 ± 140.4	32 [10–17]	-	284 [10–20]
Fixed Angle	69.7 ± 23.9	-	109.7 ± 36.3	37 [10–19]	-	38 [10–20]
Fixed Angle*	135.6 ± 76.5	-	245.2 ± 151.7	36 [10–18]	-	37 [10–20]
Interp	123.6 ± 64.5	-	292.7 ± 140.4	32 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	135.0 ± 75.3	-	291.7 ± 140.8	37 [10–19]	-	280 [10–20]
TQA*	128.9 ± 66.1	-	293.4 ± 140.4	35 [10–18]	-	280 [10–20]
Recursive TS	-	-	-	-	-	-

Table 33: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

12 Tables for depth $P = 2$

	ER	HH	RR
MPS			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	10 [40]
Fixed Angle*	-	-	10 [40]
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA*	10 [40]	10 [39]	10 [40]
PP			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	10 [40]
Fixed Angle*	-	-	10 [40]
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA*	10 [40]	10 [39]	10 [40]
SV			
Fourier	31 [10–17]	-	284 [10–20]
Fixed Angle	37 [10–19]	-	38 [10–20]
Fixed Angle*	33 [10–18]	-	37 [10–20]
Interp	31 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-
TQA	36 [10–18]	-	280 [10–20]
TQA*	31 [10–17]	-	280 [10–20]
Recursive TS	-	-	-

Table 34: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
MPS			
Fourier	471.0 ± 43.8	93.9 ± 0.0	166.9 ± 0.2
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	69.1 ± 1.3
Fixed Angle*	-	-	114.1 ± 3.6
Fourier (Aer)	-	-	-
Interp	431.8 ± 54.5	51.1 ± 0.1	166.9 ± 0.2
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	325.2 ± 21.9	70.1 ± 0.0	146.4 ± 0.3
TQA*	452.4 ± 43.2	93.9 ± 0.0	166.9 ± 0.2
PP			
Fourier	493.7 ± 44.7	91.7 ± 0.0	166.6 ± 0.2
Fixed Angle	-	-	63.2 ± 0.6
Fixed Angle*	-	-	122.7 ± 0.6
Interp	510.6 ± 61.5	91.7 ± 0.0	166.6 ± 0.1
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	353.8 ± 42.3	70.1 ± 0.0	145.8 ± 0.1
TQA*	447.3 ± 62.7	91.7 ± 0.0	166.6 ± 0.2
SV			
Fourier	169.8 ± 66.8	-	349.0 ± 128.9
Fixed Angle	94.4 ± 38.0	-	187.9 ± 133.0
Fixed Angle*	164.5 ± 57.6	-	272.7 ± 133.6
Interp	145.9 ± 79.3	-	330.9 ± 122.3
Linear Ramp*	-	-	-
TQA	146.6 ± 52.9	-	281.2 ± 81.7
TQA*	151.1 ± 45.6	-	337.0 ± 118.8
Recursive TS	-	-	-

Table 35: **Avg. Normalized Energy $\pm$ standard deviation for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	471.0 ± 43.8	93.9 ± 0.0	166.9 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	69.1 ± 1.3	-	-	10 [40]
Fixed Angle*	-	-	114.1 ± 3.6	-	-	10 [40]
Fourier (Aer)	-	-	-	-	-	-
Interp	431.8 ± 54.5	51.1 ± 0.1	166.9 ± 0.2	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	325.2 ± 21.9	70.1 ± 0.0	146.4 ± 0.3	10 [40]	10 [39]	10 [40]
TQA*	452.4 ± 43.2	93.9 ± 0.0	166.9 ± 0.2	10 [40]	10 [39]	10 [40]
PP						
Fourier	493.7 ± 44.7	91.7 ± 0.0	166.6 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	63.2 ± 0.6	-	-	10 [40]
Fixed Angle*	-	-	122.7 ± 0.6	-	-	10 [40]
Interp	510.6 ± 61.5	91.7 ± 0.0	166.6 ± 0.1	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	353.8 ± 42.3	70.1 ± 0.0	145.8 ± 0.1	10 [40]	10 [39]	10 [40]
TQA*	447.3 ± 62.7	91.7 ± 0.0	166.6 ± 0.2	10 [40]	10 [39]	10 [40]
SV						
Fourier	169.8 ± 66.8	-	349.0 ± 128.9	31 [10–17]	-	284 [10–20]
Fixed Angle	94.4 ± 38.0	-	187.9 ± 133.0	37 [10–19]	-	38 [10–20]
Fixed Angle*	164.5 ± 57.6	-	272.7 ± 133.6	33 [10–18]	-	37 [10–20]
Interp	145.9 ± 79.3	-	330.9 ± 122.3	31 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	146.6 ± 52.9	-	281.2 ± 81.7	36 [10–18]	-	280 [10–20]
TQA*	151.1 ± 45.6	-	337.0 ± 118.8	31 [10–17]	-	280 [10–20]
Recursive TS	-	-	-	-	-	-

Table 36: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P=2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

13 Tables for depth $P = 3$

	ER	HH	RR
MPS			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	-	-	-
TQA*	-	-	-
PP			
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
SV			
Fourier	31 [10–17]	-	284 [10–20]
Fixed Angle	37 [10–19]	-	34 [10–20]
Fixed Angle*	31 [10–17]	-	34 [10–20]
Interp	31 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-
TQA	35 [10–18]	-	280 [10–20]
TQA*	30 [10–17]	-	280 [10–20]
Recursive TS	-	-	-

Table 37: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
	MPS		
Fourier	484.3 ± 44.7	105.2 ± 0.1	173.8 ± 0.1
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Fourier (Aer)	-	-	-
Interp	354.0 ± 130.9	59.8 ± 1.1	173.6 ± 0.2
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	-	-	-
TQA*	-	-	-
	PP		
Fourier	546.6 ± 48.9	106.0 ± 0.0	176.2 ± 0.1
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Interp	486.2 ± 70.7	106.0 ± 0.0	176.0 ± 0.1
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
	SV		
Fourier	183.2 ± 67.2	-	368.3 ± 138.1
Fixed Angle	86.1 ± 26.4	-	119.4 ± 64.0
Fixed Angle*	181.6 ± 70.4	-	284.9 ± 113.0
Interp	146.7 ± 73.3	-	238.3 ± 82.0
Linear Ramp*	-	-	-
TQA	136.5 ± 42.3	-	280.3 ± 73.1
TQA*	180.4 ± 63.9	-	358.6 ± 117.2
Recursive TS	-	-	-

Table 38: **Avg. Normalized Energy $\pm$ standard deviation for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	484.3 ± 44.7	105.2 ± 0.1	173.8 ± 0.1	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Fourier (Aer)	-	-	-	-	-	-
Interp	354.0 ± 130.9	59.8 ± 1.1	173.6 ± 0.2	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
PP						
Fourier	546.6 ± 48.9	106.0 ± 0.0	176.2 ± 0.1	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Interp	486.2 ± 70.7	106.0 ± 0.0	176.0 ± 0.1	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fourier	183.2 ± 67.2	-	368.3 ± 138.1	31 [10–17]	-	284 [10–20]
Fixed Angle	86.1 ± 26.4	-	119.4 ± 64.0	37 [10–19]	-	34 [10–20]
Fixed Angle*	181.6 ± 70.4	-	284.9 ± 113.0	31 [10–17]	-	34 [10–20]
Interp	146.7 ± 73.3	-	238.3 ± 82.0	31 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	136.5 ± 42.3	-	280.3 ± 73.1	35 [10–18]	-	280 [10–20]
TQA*	180.4 ± 63.9	-	358.6 ± 117.2	30 [10–17]	-	280 [10–20]
Recursive TS	-	-	-	-	-	-

Table 39: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

14 Tables for depth $P = 4$

	ER	HH	RR
	MPS		
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)⋆	-	-	-
Fixed Angle	-	-	10 [40]
Fixed Angle⋆	-	-	10 [40]
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)⋆	-	-	-
Linear Ramp⋆	-	-	-
Recursive TS	-	-	-
TQA (Aer)⋆	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA⋆	10 [40]	10 [39]	10 [40]
	PP		
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	10 [40]
Fixed Angle⋆	-	-	10 [40]
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp⋆	-	-	-
Recursive TS	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA⋆	10 [40]	10 [39]	10 [40]
	SV		
Fourier	30 [10–17]	-	284 [10–20]
Fixed Angle	20 [10–17]	-	13 [10–20]
Fixed Angle⋆	20 [10–17]	-	13 [10–20]
Interp	30 [10–17]	-	284 [10–20]
Linear Ramp⋆	-	-	-
TQA	35 [10–18]	-	280 [10–20]
TQA⋆	29 [10–17]	-	280 [10–20]
Recursive TS	-	-	-

Table 40: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
MPS			
Fourier	512.3 ± 49.7	107.8 ± 0.2	174.3 ± 0.3
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	99.9 ± 5.0
Fixed Angle*	-	-	168.4 ± 0.4
Fourier (Aer)	-	-	-
Interp	331.2 ± 164.0	66.2 ± 1.8	173.8 ± 0.3
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	166.0 ± 89.5	102.7 ± 0.0	164.3 ± 0.2
TQA*	349.2 ± 129.0	106.0 ± 0.0	173.6 ± 0.2
PP			
Fourier	563.8 ± 51.5	109.3 ± 0.0	177.6 ± 0.3
Fixed Angle	-	-	65.8 ± 0.6
Fixed Angle*	-	-	165.8 ± 0.2
Interp	467.1 ± 100.1	105.4 ± 0.0	177.4 ± 0.3
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	153.5 ± 135.5	102.1 ± 0.0	162.9 ± 0.1
TQA*	371.5 ± 132.7	109.1 ± 0.0	176.4 ± 0.2
SV			
Fourier	189.6 ± 68.2	-	375.2 ± 137.4
Fixed Angle	71.2 ± 16.7	-	76.3 ± 18.5
Fixed Angle*	145.3 ± 33.8	-	167.9 ± 3.2
Interp	154.3 ± 74.7	-	233.3 ± 102.3
Linear Ramp*	-	-	-
TQA	127.6 ± 41.1	-	186.2 ± 38.4
TQA*	187.8 ± 66.6	-	333.5 ± 99.6
Recursive TS	-	-	-

Table 41: **Avg. Normalized Energy $\pm$ standard deviation for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	512.3 ± 49.7	107.8 ± 0.2	174.3 ± 0.3	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	99.9 ± 5.0	-	-	10 [40]
Fixed Angle*	-	-	168.4 ± 0.4	-	-	10 [40]
Fourier (Aer)	-	-	-	-	-	-
Interp	331.2 ± 164.0	66.2 ± 1.8	173.8 ± 0.3	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	166.0 ± 89.5	102.7 ± 0.0	164.3 ± 0.2	10 [40]	10 [39]	10 [40]
TQA*	349.2 ± 129.0	106.0 ± 0.0	173.6 ± 0.2	10 [40]	10 [39]	10 [40]
PP						
Fourier	563.8 ± 51.5	109.3 ± 0.0	177.6 ± 0.3	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	65.8 ± 0.6	-	-	10 [40]
Fixed Angle*	-	-	165.8 ± 0.2	-	-	10 [40]
Interp	467.1 ± 100.1	105.4 ± 0.0	177.4 ± 0.3	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	153.5 ± 135.5	102.1 ± 0.0	162.9 ± 0.1	10 [40]	10 [39]	10 [40]
TQA*	371.5 ± 132.7	109.1 ± 0.0	176.4 ± 0.2	10 [40]	10 [39]	10 [40]
SV						
Fourier	189.6 ± 68.2	-	375.2 ± 137.4	30 [10–17]	-	284 [10–20]
Fixed Angle	71.2 ± 16.7	-	76.3 ± 18.5	20 [10–17]	-	13 [10–20]
Fixed Angle*	145.3 ± 33.8	-	167.9 ± 3.2	20 [10–17]	-	13 [10–20]
Interp	154.3 ± 74.7	-	233.3 ± 102.3	30 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	127.6 ± 41.1	-	186.2 ± 38.4	35 [10–18]	-	280 [10–20]
TQA*	187.8 ± 66.6	-	333.5 ± 99.6	29 [10–17]	-	280 [10–20]
Recursive TS	-	-	-	-	-	-

Table 42: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P=4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

15 Tables for depth $P = 5$

	ER	HH	RR
		MPS	
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	-	-	-
TQA*	-	-	-
		PP	
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-
Fixed Angle*	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA*	-	-	-
		SV	
Fourier	28 [10–17]	-	284 [10–20]
Fixed Angle	11 [10–15]	-	10 [20]
Fixed Angle*	11 [10–15]	-	10 [20]
Interp	28 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-
TQA	35 [10–18]	-	284 [10–20]
TQA*	27 [10–16]	-	283 [10–20]
Recursive TS	-	-	-

Table 43: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
		MPS	
Fourier	535.9 ± 52.0	110.2 ± 0.1	174.6 ± 0.1
Fixed Angle (Aer)⋆	-	-	-
Fixed Angle	-	-	-
Fixed Angle⋆	-	-	-
Fourier (Aer)	-	-	-
Interp	317.9 ± 182.7	68.0 ± 0.8	174.7 ± 0.1
Interp (Aer)	-	-	-
Linear Ramp (Aer)⋆	-	-	-
Linear Ramp⋆	-	-	-
Recursive TS	-	-	-
TQA (Aer)⋆	-	-	-
TQA	-	-	-
TQA⋆	-	-	-
		PP	
Fourier	566.9 ± 51.4	110.1 ± 0.0	179.2 ± 0.2
Fixed Angle	-	-	-
Fixed Angle⋆	-	-	-
Interp	404.2 ± 123.5	105.1 ± 0.0	179.1 ± 0.2
Linear Ramp⋆	-	-	-
Recursive TS	-	-	-
TQA	-	-	-
TQA⋆	-	-	-
		SV	
Fourier	187.9 ± 64.0	-	379.1 ± 141.1
Fixed Angle	68.7 ± 15.2	-	56.5 ± 0.5
Fixed Angle⋆	125.0 ± 17.9	-	159.2 ± 8.8
Interp	144.7 ± 67.5	-	220.0 ± 102.8
Linear Ramp⋆	-	-	-
TQA	130.2 ± 42.0	-	193.1 ± 39.1
TQA⋆	178.4 ± 54.5	-	346.4 ± 111.3
Recursive TS	-	-	-

Table 44: **Avg. Normalized Energy** $\pm$ **standard deviation for** $P = 5$**.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	535.9 ± 52.0	110.2 ± 0.1	174.6 ± 0.1	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Fourier (Aer)	-	-	-	-	-	-
Interp	317.9 ± 182.7	68.0 ± 0.8	174.7 ± 0.1	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
PP						
Fourier	566.9 ± 51.4	110.1 ± 0.0	179.2 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	-	-	-	-
Fixed Angle*	-	-	-	-	-	-
Interp	404.2 ± 123.5	105.1 ± 0.0	179.1 ± 0.2	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	-	-	-	-	-	-
TQA*	-	-	-	-	-	-
SV						
Fourier	187.9 ± 64.0	-	379.1 ± 141.1	28 [10–17]	-	284 [10–20]
Fixed Angle	68.7 ± 15.2	-	56.5 ± 0.5	11 [10–15]	-	10 [20]
Fixed Angle*	125.0 ± 17.9	-	159.2 ± 8.8	11 [10–15]	-	10 [20]
Interp	144.7 ± 67.5	-	220.0 ± 102.8	28 [10–17]	-	284 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	130.2 ± 42.0	-	193.1 ± 39.1	35 [10–18]	-	284 [10–20]
TQA*	178.4 ± 54.5	-	346.4 ± 111.3	27 [10–16]	-	283 [10–20]
Recursive TS	-	-	-	-	-	-

Table 45: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

16 Tables for depth $P = 6$

	ER	HH	RR
	MPS		
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	10 [40]
Fixed Angle*	-	-	10 [40]
Fourier (Aer)	-	-	-
Interp	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA*	10 [40]	10 [39]	10 [40]
	PP		
Fourier	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	10 [40]
Fixed Angle*	-	-	10 [40]
Interp	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	10 [40]	10 [39]	10 [40]
TQA*	10 [40]	10 [39]	10 [40]
	SV		
Fourier	-	-	41 [10–20]
Fixed Angle	-	-	10 [20]
Fixed Angle*	-	-	10 [20]
Interp	-	-	40 [10–20]
Linear Ramp*	-	-	-
TQA	-	-	41 [10–20]
TQA*	-	-	40 [10–20]
Recursive TS	-	-	-

Table 46: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	ER	HH	RR
MPS			
Fourier	539.1 ± 47.5	110.4 ± 0.1	174.8 ± 0.1
Fixed Angle (Aer)*	-	-	-
Fixed Angle	-	-	135.9 ± 3.0
Fixed Angle*	-	-	167.0 ± 1.0
Fourier (Aer)	-	-	-
Interp	335.3 ± 180.2	70.6 ± 2.0	174.8 ± 0.0
Interp (Aer)	-	-	-
Linear Ramp (Aer)*	-	-	-
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA (Aer)*	-	-	-
TQA	137.7 ± 81.8	102.8 ± 0.0	165.9 ± 0.1
TQA*	329.0 ± 93.3	109.3 ± 0.1	173.6 ± 0.3
PP			
Fourier	562.2 ± 50.5	110.6 ± 0.0	180.0 ± 0.2
Fixed Angle	-	-	105.2 ± 0.4
Fixed Angle*	-	-	171.8 ± 0.9
Interp	467.6 ± 114.7	107.6 ± 0.0	179.4 ± 0.2
Linear Ramp*	-	-	-
Recursive TS	-	-	-
TQA	75.5 ± 78.8	103.9 ± 0.0	168.4 ± 0.1
TQA*	274.4 ± 139.6	109.9 ± 0.0	177.9 ± 0.6
SV			
Fourier	-	-	328.1 ± 149.0
Fixed Angle	-	-	103.7 ± 1.4
Fixed Angle*	-	-	169.5 ± 2.0
Interp	-	-	200.6 ± 94.3
Linear Ramp*	-	-	-
TQA	-	-	173.6 ± 61.4
TQA*	-	-	289.5 ± 115.2
Recursive TS	-	-	-

Table 47: **Avg. Normalized Energy $\pm$ standard deviation for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy			Num. Instances		
	ER	HH	RR	ER	HH	RR
MPS						
Fourier	539.1 ± 47.5	110.4 ± 0.1	174.8 ± 0.1	10 [40]	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-	-	-
Fixed Angle	-	-	135.9 ± 3.0	-	-	10 [40]
Fixed Angle*	-	-	167.0 ± 1.0	-	-	10 [40]
Fourier (Aer)	-	-	-	-	-	-
Interp	335.3 ± 180.2	70.6 ± 2.0	174.8 ± 0.0	10 [40]	10 [39]	10 [40]
Interp (Aer)	-	-	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-	-	-
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA (Aer)*	-	-	-	-	-	-
TQA	137.7 ± 81.8	102.8 ± 0.0	165.9 ± 0.1	10 [40]	10 [39]	10 [40]
TQA*	329.0 ± 93.3	109.3 ± 0.1	173.6 ± 0.3	10 [40]	10 [39]	10 [40]
PP						
Fourier	562.2 ± 50.5	110.6 ± 0.0	180.0 ± 0.2	10 [40]	10 [39]	10 [40]
Fixed Angle	-	-	105.2 ± 0.4	-	-	10 [40]
Fixed Angle*	-	-	171.8 ± 0.9	-	-	10 [40]
Interp	467.6 ± 114.7	107.6 ± 0.0	179.4 ± 0.2	10 [40]	10 [39]	10 [40]
Linear Ramp*	-	-	-	-	-	-
Recursive TS	-	-	-	-	-	-
TQA	75.5 ± 78.8	103.9 ± 0.0	168.4 ± 0.1	10 [40]	10 [39]	10 [40]
TQA*	274.4 ± 139.6	109.9 ± 0.0	177.9 ± 0.6	10 [40]	10 [39]	10 [40]
SV						
Fourier	-	-	328.1 ± 149.0	-	-	41 [10–20]
Fixed Angle	-	-	103.7 ± 1.4	-	-	10 [20]
Fixed Angle*	-	-	169.5 ± 2.0	-	-	10 [20]
Interp	-	-	200.6 ± 94.3	-	-	40 [10–20]
Linear Ramp*	-	-	-	-	-	-
TQA	-	-	173.6 ± 61.4	-	-	41 [10–20]
TQA*	-	-	289.5 ± 115.2	-	-	40 [10–20]
Recursive TS	-	-	-	-	-	-

Table 48: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

17 Tables for depth $P = 7$

	HH	RR
MPS		
Fourier	10 [39]	10 [40]
Fixed Angle (Aer)⋆	-	-
Fixed Angle	-	-
Fixed Angle⋆	-	-
Fourier (Aer)	-	-
Interp	10 [39]	10 [40]
Interp (Aer)	-	-
Linear Ramp (Aer)⋆	-	-
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA (Aer)⋆	-	-
TQA	-	-
TQA⋆	-	-
PP		
Fourier	10 [39]	10 [40]
Fixed Angle	-	-
Fixed Angle⋆	-	-
Interp	10 [39]	10 [40]
Linear Ramp⋆	-	-
Recursive TS	-	-
TQA	-	-
TQA⋆	-	-
SV		
Fourier	-	41 [10–20]
Fixed Angle	-	10 [20]
Fixed Angle⋆	-	10 [20]
Interp	-	40 [10–20]
Linear Ramp⋆	-	-
TQA	-	41 [10–20]
TQA⋆	-	40 [10–20]
Recursive TS	-	-

Table 49: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	HH	RR
MPS		
Fourier	110.8 ± 0.1	174.8 ± 0.1
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	75.5 ± 1.9	174.8 ± 0.0
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	110.8 ± 0.0	179.7 ± 0.3
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	107.3 ± 0.0	179.9 ± 0.4
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	-	322.9 ± 145.7
Fixed Angle	-	83.3 ± 0.5
Fixed Angle*	-	161.9 ± 2.3
Interp	-	196.8 ± 103.7
Linear Ramp*	-	-
TQA	-	173.2 ± 56.1
TQA*	-	271.7 ± 99.1
Recursive TS	-	-

Table 50: **Avg. Normalized Energy $\pm$ standard deviation for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy		Num. Instances	
	HH	RR	HH	RR
MPS				
Fourier	110.8 ± 0.1	174.8 ± 0.1	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Fourier (Aer)	-	-	-	-
Interp	75.5 ± 1.9	174.8 ± 0.0	10 [39]	10 [40]
Interp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
PP				
Fourier	110.8 ± 0.0	179.7 ± 0.3	10 [39]	10 [40]
Fixed Angle	-	-	-	-
Fixed Angle*	-	-	-	-
Interp	107.3 ± 0.0	179.9 ± 0.4	10 [39]	10 [40]
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	-	-	-	-
TQA*	-	-	-	-
SV				
Fourier	-	322.9 ± 145.7	-	41 [10–20]
Fixed Angle	-	83.3 ± 0.5	-	10 [20]
Fixed Angle*	-	161.9 ± 2.3	-	10 [20]
Interp	-	196.8 ± 103.7	-	40 [10–20]
Linear Ramp*	-	-	-	-
TQA	-	173.2 ± 56.1	-	41 [10–20]
TQA*	-	271.7 ± 99.1	-	40 [10–20]
Recursive TS	-	-	-	-

Table 51: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

18 Tables for depth $P = 8$

	HH	RR
MPS		
Fourier	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-
Fixed Angle	-	10 [40]
Fixed Angle*	-	10 [40]
Fourier (Aer)	-	-
Interp	10 [39]	10 [40]
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	10 [39]	10 [40]
TQA*	10 [39]	10 [40]
PP		
Fourier	10 [39]	10 [40]
Fixed Angle	-	10 [40]
Fixed Angle*	-	10 [40]
Interp	10 [39]	10 [40]
Linear Ramp*	-	-
Recursive TS	-	-
TQA	10 [39]	10 [40]
TQA*	10 [39]	10 [40]
SV		
Fourier	-	41 [10–20]
Fixed Angle	-	10 [20]
Fixed Angle*	-	10 [20]
Interp	-	40 [10–20]
Linear Ramp*	-	-
TQA	-	41 [10–20]
TQA*	-	40 [10–20]
Recursive TS	-	-

Table 52: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	HH	RR
MPS		
Fourier	110.8 ± 0.1	174.8 ± 0.1
Fixed Angle (Aer)*	-	-
Fixed Angle	-	132.5 ± 2.0
Fixed Angle*	-	165.5 ± 1.7
Fourier (Aer)	-	-
Interp	80.6 ± 2.4	174.8 ± 0.1
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	106.0 ± 0.0	167.6 ± 0.1
TQA*	110.0 ± 0.0	172.3 ± 0.2
PP		
Fourier	110.7 ± 0.0	179.2 ± 0.3
Fixed Angle	-	90.6 ± 0.6
Fixed Angle*	-	174.9 ± 1.1
Interp	108.1 ± 0.0	179.5 ± 0.4
Linear Ramp*	-	-
Recursive TS	-	-
TQA	108.3 ± 0.0	170.2 ± 0.1
TQA*	110.1 ± 0.0	177.4 ± 0.4
SV		
Fourier	-	325.1 ± 148.7
Fixed Angle	-	102.1 ± 1.6
Fixed Angle*	-	162.7 ± 22.0
Interp	-	189.5 ± 104.1
Linear Ramp*	-	-
TQA	-	173.1 ± 55.1
TQA*	-	279.3 ± 107.4
Recursive TS	-	-

Table 53: **Avg. Normalized Energy $\pm$ standard deviation for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy		Num. Instances	
	HH	RR	HH	RR
MPS				
Fourier	110.8 ± 0.1	174.8 ± 0.1	10 [39]	10 [40]
Fixed Angle (Aer)*	-	-	-	-
Fixed Angle	-	132.5 ± 2.0	-	10 [40]
Fixed Angle*	-	165.5 ± 1.7	-	10 [40]
Fourier (Aer)	-	-	-	-
Interp	80.6 ± 2.4	174.8 ± 0.1	10 [39]	10 [40]
Interp (Aer)	-	-	-	-
Linear Ramp (Aer)*	-	-	-	-
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA (Aer)*	-	-	-	-
TQA	106.0 ± 0.0	167.6 ± 0.1	10 [39]	10 [40]
TQA*	110.0 ± 0.0	172.3 ± 0.2	10 [39]	10 [40]
PP				
Fourier	110.7 ± 0.0	179.2 ± 0.3	10 [39]	10 [40]
Fixed Angle	-	90.6 ± 0.6	-	10 [40]
Fixed Angle*	-	174.9 ± 1.1	-	10 [40]
Interp	108.1 ± 0.0	179.5 ± 0.4	10 [39]	10 [40]
Linear Ramp*	-	-	-	-
Recursive TS	-	-	-	-
TQA	108.3 ± 0.0	170.2 ± 0.1	10 [39]	10 [40]
TQA*	110.1 ± 0.0	177.4 ± 0.4	10 [39]	10 [40]
SV				
Fourier	-	325.1 ± 148.7	-	41 [10–20]
Fixed Angle	-	102.1 ± 1.6	-	10 [20]
Fixed Angle*	-	162.7 ± 22.0	-	10 [20]
Interp	-	189.5 ± 104.1	-	40 [10–20]
Linear Ramp*	-	-	-	-
TQA	-	173.1 ± 55.1	-	41 [10–20]
TQA*	-	279.3 ± 107.4	-	40 [10–20]
Recursive TS	-	-	-	-

Table 54: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P =$ 8.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with ± their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

19 Tables for depth $P = 9$

	RR
MPS	
Fourier	10 [40]
Fixed Angle (Aer)⋆	-
Fixed Angle	-
Fixed Angle⋆	-
Fourier (Aer)	-
Interp	10 [40]
Interp (Aer)	-
Linear Ramp (Aer)⋆	-
Linear Ramp⋆	-
Recursive TS	-
TQA (Aer)⋆	-
TQA	-
TQA⋆	-
PP	
Fourier	10 [40]
Fixed Angle	-
Fixed Angle⋆	-
Interp	10 [40]
Linear Ramp⋆	-
Recursive TS	-
TQA	-
TQA⋆	-
SV	
Fourier	40 [10–20]
Fixed Angle	10 [20]
Fixed Angle⋆	10 [20]
Interp	40 [10–20]
Linear Ramp⋆	-
TQA	41 [10–20]
TQA⋆	40 [10–20]
Recursive TS	-

Table 55: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	174.8 ± 0.1
Fixed Angle (Aer)*	-
Fixed Angle	-
Fixed Angle*	-
Fourier (Aer)	-
Interp	174.8 ± 0.1
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	-
TQA*	-
PP	
Fourier	179.0 ± 0.7
Fixed Angle	-
Fixed Angle*	-
Interp	179.3 ± 0.5
Linear Ramp*	-
Recursive TS	-
TQA	-
TQA*	-
SV	
Fourier	330.9 ± 149.7
Fixed Angle	58.1 ± 1.1
Fixed Angle*	151.6 ± 32.8
Interp	179.5 ± 106.9
Linear Ramp*	-
TQA	166.0 ± 63.0
TQA*	276.5 ± 112.1
Recursive TS	-

Table 56: **Avg. Normalized Energy $\pm$ standard deviation for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	174.8 ± 0.1	10 [40]
Fixed Angle (Aer)*	-	-
Fixed Angle	-	-
Fixed Angle*	-	-
Fourier (Aer)	-	-
Interp	174.8 ± 0.1	10 [40]
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	-	-
TQA*	-	-
PP		
Fourier	179.0 ± 0.7	10 [40]
Fixed Angle	-	-
Fixed Angle*	-	-
Interp	179.3 ± 0.5	10 [40]
Linear Ramp*	-	-
Recursive TS	-	-
TQA	-	-
TQA*	-	-
SV		
Fourier	330.9 ± 149.7	40 [10–20]
Fixed Angle	58.1 ± 1.1	10 [20]
Fixed Angle*	151.6 ± 32.8	10 [20]
Interp	179.5 ± 106.9	40 [10–20]
Linear Ramp*	-	-
TQA	166.0 ± 63.0	41 [10–20]
TQA*	276.5 ± 112.1	40 [10–20]
Recursive TS	-	-

Table 57: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.

20 Tables for depth $P = 10$

	RR
MPS	
Fourier	10 [40]
Fixed Angle (Aer)*	-
Fixed Angle	10 [40]
Fixed Angle*	10 [40]
Fourier (Aer)	-
Interp	10 [40]
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	10 [40]
TQA*	10 [40]
PP	
Fourier	10 [40]
Fixed Angle	10 [40]
Fixed Angle*	10 [40]
Interp	10 [40]
Linear Ramp*	-
Recursive TS	-
TQA	10 [40]
TQA*	10 [40]
SV	
Fourier	40 [10–20]
Fixed Angle	10 [20]
Fixed Angle*	10 [20]
Interp	39 [10–20]
Linear Ramp*	-
TQA	41 [10–20]
TQA*	40 [10–20]
Recursive TS	-

Table 58: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR). Cells are colored based on the number of instances for the given configuration, with darker colors indicating more instances.

	RR
MPS	
Fourier	174.7 ± 0.1
Fixed Angle (Aer)*	-
Fixed Angle	138.3 ± 3.9
Fixed Angle*	165.6 ± 1.3
Fourier (Aer)	-
Interp	174.8 ± 0.1
Interp (Aer)	-
Linear Ramp (Aer)*	-
Linear Ramp*	-
Recursive TS	-
TQA (Aer)*	-
TQA	164.7 ± 0.1
TQA*	172.0 ± 0.2
PP	
Fourier	178.6 ± 0.6
Fixed Angle	101.5 ± 0.7
Fixed Angle*	175.4 ± 1.6
Interp	179.1 ± 0.7
Linear Ramp*	-
Recursive TS	-
TQA	171.7 ± 0.2
TQA*	177.4 ± 0.2
SV	
Fourier	330.0 ± 150.8
Fixed Angle	100.7 ± 1.3
Fixed Angle*	169.1 ± 1.0
Interp	185.1 ± 99.1
Linear Ramp*	-
TQA	163.6 ± 63.9
TQA*	253.1 ± 88.1
Recursive TS	-

Table 59: **Avg. Normalized Energy $\pm$ standard deviation for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors, with darker colors indicating better energies. Note that energies include invalid solutions, which are penalized by the Hamiltonian.

	Energy RR	Num. Instances RR
MPS		
Fourier	174.7 ± 0.1	10 [40]
Fixed Angle (Aer)*	-	-
Fixed Angle	138.3 ± 3.9	10 [40]
Fixed Angle*	165.6 ± 1.3	10 [40]
Fourier (Aer)	-	-
Interp	174.8 ± 0.1	10 [40]
Interp (Aer)	-	-
Linear Ramp (Aer)*	-	-
Linear Ramp*	-	-
Recursive TS	-	-
TQA (Aer)*	-	-
TQA	164.7 ± 0.1	10 [40]
TQA*	172.0 ± 0.2	10 [40]
PP		
Fourier	178.6 ± 0.6	10 [40]
Fixed Angle	101.5 ± 0.7	10 [40]
Fixed Angle*	175.4 ± 1.6	10 [40]
Interp	179.1 ± 0.7	10 [40]
Linear Ramp*	-	-
Recursive TS	-	-
TQA	171.7 ± 0.2	10 [40]
TQA*	177.4 ± 0.2	10 [40]
SV		
Fourier	330.0 ± 150.8	40 [10–20]
Fixed Angle	100.7 ± 1.3	10 [20]
Fixed Angle*	169.1 ± 1.0	10 [20]
Interp	185.1 ± 99.1	39 [10–20]
Linear Ramp*	-	-
TQA	163.6 ± 63.9	41 [10–20]
TQA*	253.1 ± 88.1	40 [10–20]
Recursive TS	-	-

Table 60: **Avg. Normalized Energy and Number of Instances for Various Evaluation Methods for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).Estimated energies are normalized by the number of nodes in the graphs. Evaluation methods are shown in different colors for energies. Darker colors indicating better energies or more instances. Average energies are shown, with $\pm$ their standard deviation. The range of graph sizes are shown in brackets, next to the number of instances.